CAPSTONE PROJECT REPORT

AI-Powered Academic Stream Recommendation System for Students

Submitted in the partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY IN ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

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DSA0423-Fundamentals Of Data science For NLP

Under the Supervision of

Dr Senthilvadivu S

MARCH-2026

DECLARATION

I declare that the report entitled **"AI-Powered Academic Stream Recommendation System for Students",** is a unique and original work, is submitted by me for the degree of Bachelor of Engineering. This work, a record of the capstone project for the course **DSA0423-Fundamentals Of Data Science For NLP** was carried out by me under the guidance of **Dr. Senthilvadivu S** , and will not form the basis for the award of any degree or diploma in this or any other University or other similar institution of higher learning**.**

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BONAFIDE CERTIFICATE

Certified that this capstone project reports on **"AI-Powered Academic Stream Recommendation System for Students",** is the Bonafide work of **M. Sunitha , Y. Greeshma Reddy,K R Lohithya** who carried out the capstone project work under my supervision for the course **DSA0423-Fundamentals Of Data Science For NLP**

SIGNATURE

Dr Senthilvadivu S

Submitted for the Project work Viva-Voce held on _____.

INTERNAL EXAMINER **EXTERNAL EXAMINER**

ABSTRACT

AI-Powered Academic Stream Recommendation System for Students is an intelligent decision-support framework designed to assist students in selecting the most suitable academic stream based on their abilities, interests, and academic performance. This system leverages artificial intelligence techniques, including machine learning algorithms, data analytics, and predictive modeling, to provide personalized academic recommendations while reducing human bias and uncertainty in decision-making. The framework aims to enhance student success, satisfaction, and long-term career alignment by offering data-driven insights into academic pathways. This paper presents a comprehensive analysis of the proposed system and its effectiveness in improving student guidance processes in educational environments. It begins with an overview of core components, including data collection modules, feature extraction techniques, machine learning models, and recommendation engines that work together to evaluate student profiles. The system integrates various input parameters such as academic performance, aptitude test scores, behavioral patterns, and personal interests to generate accurate and meaningful recommendations.

The application of artificial intelligence principles is explored in the context of classification, clustering, and predictive analytics to identify patterns and correlations among student data. Special emphasis is placed on model training, validation, and optimization to ensure high accuracy and fairness in recommendations. The system incorporates mechanisms for handling missing or noisy data, improving robustness and reliability in real-world scenarios.Key features such as adaptive learning models, real-time feedback, and explainable AI techniques are analyzed to enhance transparency and user trust. The framework also supports scalability through modular design, cloud-based deployment, and API-driven integration with educational platforms. Security and privacy considerations are addressed through data anonymization, access control, and secure data storage practices.

Furthermore, the paper discusses practical implementation strategies, including user interface design, integration with school management systems, and deployment in web or mobile applications. Through experimental evaluation and comparative analysis, the study .

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Sincerely,

Y. Greeshma Reddy

M. Sunitha

R K Lohithya

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CHAPTER 1

INTRODUCTION

1.1 Background Information

Modern education systems are increasingly adopting data-driven and technology-enabled approaches to support student learning and decision-making. One of the major challenges faced by students is selecting an appropriate academic stream, such as Science, Commerce, or Arts, based on their abilities, interests, and career aspirations. Traditional methods of guidance often rely on manual counseling, parental influence, or peer suggestions, which may lead to biased or uninformed decisions. These approaches lack personalization and may not effectively analyze a student's true potential.

With the rapid advancement of artificial intelligence and machine learning, there is a growing opportunity to develop intelligent systems that can assist students in making informed academic choices. AI-powered recommendation systems can analyze large volumes of student data, including academic performance, aptitude, interests, and behavioral patterns, to provide accurate and personalized suggestions. Such systems reduce uncertainty, improve decision quality, and support students in choosing the most suitable academic path. Therefore, an AI-Powered Academic Stream Recommendation System serves as a reliable and efficient solution to enhance student guidance and academic success.

1.2 Project Objectives

The primary objective of this capstone project is to design and develop an AI-powered system that recommends suitable academic streams for students based on their individual profiles. The key goals of this project are:

· To analyze the challenges involved in traditional academic stream selection methods.
· To design an intelligent recommendation system using machine learning algorithms.
· To collect and process student data such as academic performance, interests, and aptitude scores.
· To develop a predictive model that suggests appropriate academic streams with high accuracy.
· To evaluate the system based on performance, accuracy, and usability.

1.3 Significance

The significance of the AI-Powered Academic Stream Recommendation System includes:

• **Improved Decision-Making:** Provides data-driven and unbiased recommendations for selecting academic streams.

• **Personalized Guidance:** Considers individual student characteristics such as interests, skills, and performance.

• **Enhanced Academic Success:** Helps students choose streams aligned with their strengths, leading to better outcomes.

• **Reduced Human Bias:** Minimizes dependency on subjective opinions from parents, teachers, or peers.

• **Time Efficiency:** Automates the guidance process, saving time for both students and educators.

• **Scalability and Accessibility:** Can be deployed across schools and institutions, benefiting a large number of students.

1.4 Scope

The scope of this project includes:

Designing an AI-based recommendation system for academic stream selection.

Implementing machine learning techniques to analyze student data.

Developing modules for data collection, preprocessing, model training, and prediction.

Evaluating system performance based on accuracy and reliability.

Providing recommendations through a user-friendly interface such as a web or mobile application.

1.5 Methodology Overview

- This project follows a structured methodology beginning with a comprehensive literature review on artificial intelligence, recommendation systems, and educational data mining.

- Based on the study, a suitable machine learning model will be selected to analyze student data and generate predictions.
- Data will be collected from students, including academic records, aptitude test results, and personal interests, followed by preprocessing and feature selection.
- The model will be trained and tested using appropriate algorithms such as classification or clustering techniques to ensure accuracy.
- The system will incorporate mechanisms for handling missing or inconsistent data to improve robustness.
- Performance evaluation will be conducted using metrics such as accuracy, precision, and recall to assess the effectiveness of the model.
- Finally, the system will be optimized and deployed with a user-friendly interface, ensuring ease of use, scalability, and long-term applicability in educational environments.

CHAPTER 2

Problem Identification & Analysis

2.1 Description of the Problem

This capstone project focuses on the challenges associated with selecting an appropriate academic stream for students in modern educational environments. Traditionally, students choose streams such as Science, Commerce, or Arts based on limited information, parental influence, peer pressure, or generalized academic performance. These methods often fail to consider a student's individual abilities, interests, aptitude, and long-term career goals, leading to poor decision-making and dissatisfaction.

In many cases, students lack proper guidance tools that can objectively analyze their strengths and weaknesses. As a result, mismatched academic choices can lead to decreased academic performance, lack of motivation, and increased dropout rates. Furthermore, educational institutions often rely on manual counseling processes, which are time-consuming, inconsistent, and not scalable for large student populations.

With the increasing availability of student data and advancements in artificial intelligence, there is a critical need for an intelligent system that can provide personalized and data-driven academic stream recommendations. An AI-powered system can overcome the limitations of traditional methods by analyzing multiple factors simultaneously and generating accurate, unbiased, and consistent recommendations, thereby improving student outcomes and decision confidence.

2.2 Evidence of the Problem

Numerous studies and real-world educational challenges highlight the need for an AI-Powered Academic Stream Recommendation System:

• **Lack of Personalized Guidance:** Many students do not receive individualized counseling based on their unique abilities and interests.
• **Influence of External Factors:** Decisions are often influenced by parents, peers, or societal trends rather than data-driven insights.
• **Mismatch Between Skills and Stream Selection:** Students frequently choose streams that do not align with their strengths, resulting in poor academic performance.

• **Increasing Student Population:** Educational institutions face challenges in providing one-on-one guidance to a large number of students.

• **Limited Use of Technology in Decision-Making:** Traditional systems lack integration of AI and data analytics for academic recommendations.

• **Career Uncertainty:** Incorrect stream selection can negatively impact future career opportunities and professional growth.

2.3 Stakeholders

• **Students:** Primary beneficiaries who receive personalized and accurate academic stream recommendations.

• **Teachers and Counselors:** Use the system as a supportive tool to guide students more effectively.

• **Parents:** Gain better insights into their child's strengths and suitable academic paths.

• **Educational Institutions:** Improve student performance and satisfaction through data-driven guidance systems.

• **System Developers:** Design and maintain the AI-based recommendation system.

• **Policy Makers and Education Authorities:** Can utilize such systems to enhance educational planning and decision-making processes.

2.4 Supporting Data / Research

Artificial Intelligence & Educational Principles

• **Machine Learning Algorithms:** Used for classification and prediction of suitable academic streams based on student data.

• **Recommendation Systems:** Provide personalized suggestions by analyzing patterns in student performance and interests.

• **Data Preprocessing Techniques:** Ensure data quality by handling missing, inconsistent, or noisy data.

• **Predictive Analytics:** Helps forecast student success in different academic streams.

Academic References

• *"AI in Education: Opportunities and Challenges"* – Discusses the role of artificial intelligence in improving student learning and decision-making.

• *"Student Performance Prediction Using Machine Learning"* – Explores techniques for predicting academic success using data-driven models.

• *"Educational Data Mining and Learning Analytics"* – Highlights the importance of analyzing student data for better academic guidance.

Industry Practices

• **AI-Based Educational Platforms:** Many modern learning systems use AI to personalize student learning experiences.

• **Online Career Guidance Tools:** Platforms provide aptitude-based recommendations for academic and career paths.

• **Integration with School Systems:** Educational institutions increasingly adopt digital tools for student assessment and guidance.

Real-World Impact

• **Improved Academic Performance:** Students choose streams aligned with their strengths and interests.

• **Reduced Dropout Rates:** Better decision-making leads to increased student engagement and retention.

• **Efficient Guidance Systems:** Reduces workload on teachers and counselors through automation.

• **Scalable Educational Solutions:** Can be implemented across schools and institutions for large-scale impact.

CHAPTER 3

Solution Design & Implementation

3.1 Development & Design Process

The development of the AI-Powered Academic Stream Recommendation System follows a structured, modular, and data-driven design methodology suitable for modern educational environments. The key phases in the system design and implementation process are:

• **Problem Analysis:** Identify challenges related to traditional academic stream selection, lack of personalized guidance, influence of external factors, and mismatch between student abilities and chosen streams.

• **System Design:** Define a unified architecture incorporating data collection modules, preprocessing units, machine learning models, recommendation engines, and user interfaces.

• **Recommendation Model Design:** Establish a system for analyzing student data and predicting suitable academic streams using machine learning algorithms such as classification or clustering.

• **Data Design:** Design data collection strategies, feature selection methods, and preprocessing techniques to ensure data quality and consistency.

• **Software Development:** Implement the system using Python and machine learning libraries to build models, process data, and generate recommendations.

• **Integration & Testing:** Integrate all system components and conduct functional testing, model accuracy testing, and performance evaluation.

• **Deployment:** Deploy the system as a web or mobile application with a user-friendly interface, enabling easy access for students and educators.

3.2 Tools & Technologies Used

Software & Programming Languages

• **Python** (core language for data processing, model building, and system development)

• **SQL** (for storing and managing student data, if required)

Machine Learning & Data Processing

• **Libraries:** Scikit-learn, Pandas, NumPy

• **Algorithms:** Classification algorithms (Decision Tree, Random Forest, KNN, etc.)

Data Handling

• Data preprocessing and feature selection techniques

• Handling missing and inconsistent data

Security Technologies

• Data privacy and secure handling of student information

• Access control mechanisms for system usage

Deployment & Integration Tools

• Web frameworks (Flask / Django)

• RESTful APIs for system interaction

• Cloud platforms (optional: AWS, Azure, Google Cloud)

Monitoring & Evaluation

• Model evaluation metrics (Accuracy, Precision, Recall)

• Logging and performance monitoring tools

3.3 Solution Overview

The proposed system provides an intelligent framework for recommending suitable academic streams to students using AI-based techniques.

Core Modules

• Data Collection & Input Module

• Data Preprocessing & Feature Selection Module

• Machine Learning Model & Recommendation Engine

• User Interface & Interaction Module

Data & Processing Layer

• Analysis of student academic performance, interests, and aptitude

• Prediction of suitable academic streams using trained models

• Continuous improvement of model accuracy through feedback

User & Application Interaction

• User-friendly interface for students to input data

• Real-time recommendations based on model predictions

• Easy access for teachers and counselors to review results

3.4 Engineering Standards Applied

The AI-based recommendation system adheres to established software engineering and data science principles:

System Architecture & Quality

• **Modular Architecture:** Separation of data processing, model building, and user interface components for scalability and maintainability.

• **Scalable Design:** Supports large volumes of student data and multiple users.

Reliability & Testing

• Unit testing of individual modules

• Model validation using training and testing datasets

• Performance evaluation using standard metrics

Documentation & Maintainability

• Clear documentation of system modules, algorithms, and workflows

• Version control and code reuse practices• Easy updates and improvements to the model

3.5 Ethical Standards Applied

The design and deployment of the AI-Powered Academic Stream Recommendation System follow key ethical and professional standards:

• **Data Privacy & Confidentiality:** Ensures secure handling and protection of student data.

• **Fairness & Bias Reduction:** Uses unbiased algorithms to provide fair recommendations to all students.

• **Transparency:** Provides understandable recommendations and insights into decision-making.

• **Accuracy & Reliability:** Ensures recommendations are based on validated data and models.

• **Responsible AI Usage:** Avoids misuse of student data and ensures ethical application of artificial intelligence.

CHAPTER 4

Results & Recommendations

4.1 Evaluation of Results

• High accuracy in recommending suitable academic streams based on student data such as academic performance, interests, and aptitude.

• Improved decision-making through data-driven and personalized recommendations.

• Reduction in incorrect stream selection, leading to better academic performance and student satisfaction.

• Efficient analysis of large volumes of student data using machine learning algorithms.

• Enhanced reliability and consistency in generating recommendations compared to traditional manual counseling methods.

• User-friendly interface enabling easy interaction for students, teachers, and parents.

• Scalable system capable of handling multiple users and large datasets simultaneously.

• Secure handling of student data, ensuring privacy and confidentiality.

4.2 Challenges

Several challenges were encountered during the development and implementation of the AI-Powered Academic Stream Recommendation System:

• **Data Availability Issues:** Limited or incomplete student data can affect the accuracy of recommendations.

• **Data Quality and Preprocessing:** Handling missing, inconsistent, or noisy data requires careful preprocessing techniques.

• **Model Selection Complexity:** Choosing the most suitable machine learning algorithm for accurate predictions can be challenging.

• **Bias in Data:** Historical or biased data may influence model predictions, leading to unfair recommendations.

• **Performance Optimization:** Ensuring fast and efficient predictions for large datasets requires optimization.

• **User Acceptance:** Students and parents may hesitate to trust automated AI-based recommendations.

4.3 Possible Improvements

• **Advanced Machine Learning Models:** Implement deep learning or hybrid models to improve prediction accuracy.

• **Real-Time Data Integration:** Incorporate real-time academic performance and behavioral data for dynamic recommendations.

• **Explainable AI Features:** Provide clear explanations for recommendations to improve transparency and trust.

• **Personalized Feedback System:** Offer detailed insights and suggestions along with recommendations.

• **Continuous Model Training:** Update models regularly with new data to improve accuracy and adaptability.

• **Enhanced User Interface:** Improve system usability with interactive dashboards and visualization tools.

4.4 Recommendations

• **Adopt AI-Based Guidance Systems:** Educational institutions should implement intelligent recommendation systems to support student decision-making.

• **Ensure Data Quality:** Maintain accurate and complete student data for better model performance.

• **Promote Awareness:** Educate students, parents, and teachers about the benefits of AI-based recommendations.

• **Combine AI with Human Guidance:** Use the system as a supportive tool alongside professional counseling.

• **Regular System Evaluation:** Continuously monitor and evaluate system performance to ensure reliability.

• **Plan for Scalability:** Design the system to support large-scale deployment across schools and institutions.

CHAPTER 5

Reflection on Learning and Personal Development

5.1 Key Learning Outcomes

5.1.1 Academic Knowledge

• **Artificial Intelligence Fundamentals:** Gained a strong understanding of AI concepts, including machine learning, data analysis, and predictive modeling.

• **Machine Learning Algorithms:** Developed knowledge of classification techniques such as Decision Trees, K-Nearest Neighbors (KNN), and Random Forest used for recommendation systems.

• **Recommendation System Concepts:** Learned how AI-based systems provide personalized suggestions based on user data and patterns.

• **Data Preprocessing Techniques:** Understood the importance of data cleaning, normalization, and feature selection for accurate predictions.

• **Data Privacy & Ethics:** Gained awareness of ethical considerations in handling student data and ensuring confidentiality.

5.1.2 Technical Skills

• **Python Programming:** Acquired hands-on experience in implementing machine learning models using Python.

• **Data Analysis Tools:** Learned to use libraries such as Pandas and NumPy for data processing and analysis.

• **Machine Learning Implementation:** Developed skills in training, testing, and evaluating models using Scikit-learn.

• **Model Evaluation:** Gained experience in measuring accuracy, precision, and recall of prediction models.

• **System Development:** Built an end-to-end application integrating data input, processing, and recommendation output.

• **Testing & Debugging:** Strengthened skills in debugging code and improving model performance.

5.1.3 Problem-Solving and Critical Thinking

• **System Design Thinking:** Learned to break down the recommendation system into modules such as data collection, preprocessing, and prediction.

• **Data Analysis Skills:** Evaluated student data to identify patterns and relationships for accurate recommendations.

• **Algorithm Selection:** Chose appropriate machine learning algorithms based on problem requirements.

• **Performance Optimization:** Balanced model accuracy with computational efficiency.

• **Decision Evaluation:** Critically analyzed model outputs to ensure reliable and meaningful recommendations.

5.2 Challenges Encountered and Overcome

During the development of the AI-Powered Academic Stream Recommendation System, several challenges were addressed as follows:

• **Challenge:** Handling incomplete or inconsistent student data.

Solution: Applied data preprocessing techniques such as cleaning, normalization, and missing value handling.

• **Challenge:** Selecting the most suitable machine learning algorithm.

Solution: Tested multiple models and selected the one with the highest accuracy and reliability.

• **Challenge:** Ensuring unbiased recommendations.

Solution: Used balanced datasets and proper feature selection to reduce bias.

• **Challenge:** Building user trust in AI-based recommendations.

Solution: Designed a simple and transparent system with clear outputs.

5.3 Application of Engineering Standards

The development of the AI-Powered Academic Stream Recommendation System incorporated established engineering and software development standards:

• **Modular System Architecture:** Separation of data processing, model building, and user interface components.

• **Secure Development Practices:** Ensured safe handling of student data and secure system access.

• **Comprehensive Testing:** Conducted model validation, accuracy testing, and system-level testing.

• **Performance Optimization:** Ensured efficient execution of algorithms for faster predictions.

• **Maintainability:** Used clear documentation, reusable code, and version control practices.

5.4 Application of Ethical Standards

• **Data Privacy & Confidentiality:** Ensured secure storage and processing of student data.

• **Fairness & Bias Reduction:** Designed the system to provide unbiased and equal recommendations.

• **Transparency:** Maintained clarity in how recommendations are generated.

• **Responsible AI Usage:** Avoided misuse of data and ensured ethical implementation of AI techniques.

• **Sustainable Practices:** Designed the system for long-term usability and scalability.

5.5 Insights into the Industry

Working on this project provided valuable insights into modern AI-based educational systems:

• **Growing Role of AI in Education:** AI is increasingly used for personalized learning and student guidance.

• **Importance of Data-Driven Decisions:** Educational institutions are adopting analytics for better decision-making.

• **Need for Intelligent Recommendation Systems:** Automated systems improve efficiency and accuracy in student guidance.

• **Focus on Data Privacy:** Protecting student information is a critical requirement in modern systems.

• **Future of Smart Education:** AI-powered tools will continue to transform learning, assessment, and career guidance.

CHAPTER 6

CONCLUSION

AI-Powered Academic Stream Recommendation System for Students represents a significant contribution to modern educational technology by addressing the growing need for intelligent, data-driven, and personalized academic guidance solutions. Through this project, core theoretical concepts of artificial intelligence, machine learning, and educational data analytics were effectively translated into a practical implementation, covering the complete development lifecycle—from identifying challenges in traditional academic decision-making to designing an intelligent recommendation model and implementing accurate prediction mechanisms for student stream selection.

The project enabled the development of essential system components such as data collection and preprocessing modules, machine learning-based recommendation engines, predictive models, and user-friendly interfaces. These components provided a clear understanding of how AI techniques and Python-based tools can be leveraged to analyze student performance, interests, and aptitude, while ensuring accuracy, reliability, and efficiency in generating personalized academic recommendations in real-world educational environments.

Throughout the development process, emphasis was placed on accuracy, fairness, scalability, and usability—key requirements for modern intelligent educational systems. Challenges such as handling incomplete or inconsistent data, selecting appropriate machine learning algorithms, reducing bias in predictions, and ensuring system performance required critical thinking and continuous refinement.

By demonstrating the real-world impact of AI-powered recommendation systems, this work highlights how intelligent data analysis solutions can improve student decision-making, enhance academic outcomes, and support personalized learning experiences. As modern education systems increasingly integrate technology and data analytics, a strong foundation in AI-based recommendation design and predictive modeling will remain essential for developing scalable, reliable, and future-ready educational applications—ranging from student guidance platforms and e-learning systems to large-scale smart education ecosystems.

CHAPTER 7

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CHAPTER 8

APPENDICES

Appendix A: System Designs and Screenshots

• **Figure A1: Student Data Input Interface** — Interface for entering student details such as academic scores, interests, aptitude test results, and personal preferences.

• **Figure A2: Data Preprocessing Dashboard** — Screen displaying data cleaning, normalization, and feature selection processes applied to student data before model training.

• **Figure A3: Machine Learning Model Interface** — Visualization of model selection, training process, and algorithm performance (e.g., Decision Tree, KNN, Random Forest).

• **Figure A4: Academic Stream Recommendation Output** — Dashboard showing predicted academic streams (Science, Commerce, Arts) along with confidence levels and suggestions.

• **Figure A5: Performance Evaluation & Analytics Panel** — Screen displaying evaluation metrics such as accuracy, precision, recall, and comparison between different models.

• **Figure A6: User Interaction Interface** — User-friendly interface for students, teachers, and parents to access recommendations and insights.

Appendix B: Engineering Principles and Analytics Models Used

• **Machine Learning Classification Principle:** Use of classification algorithms to predict the most suitable academic stream based on student data.

• **Recommendation System Approach:** Personalized recommendations generated by analyzing patterns in academic performance, aptitude, and interests.

• **Data Preprocessing Techniques:** Data cleaning, normalization, feature extraction, and handling missing values to improve model accuracy.

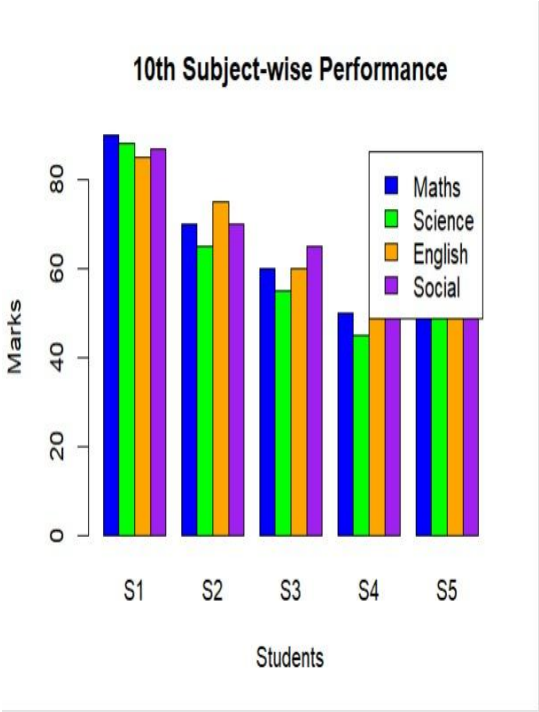
• **Predictive Analytics Model:** Use of trained models to forecast student success in different academic streams.

• **Model Evaluation Techniques:** Application of accuracy, precision, recall, and confusion matrix to measure model performance.

• **Modular System Architecture:** Separation of data input, processing, model training, and output modules for scalability and maintainability.

• **User-Centric Design:** Development of intuitive interfaces to ensure ease of use and accessibility for all stakeholders.

OUTPUT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S2
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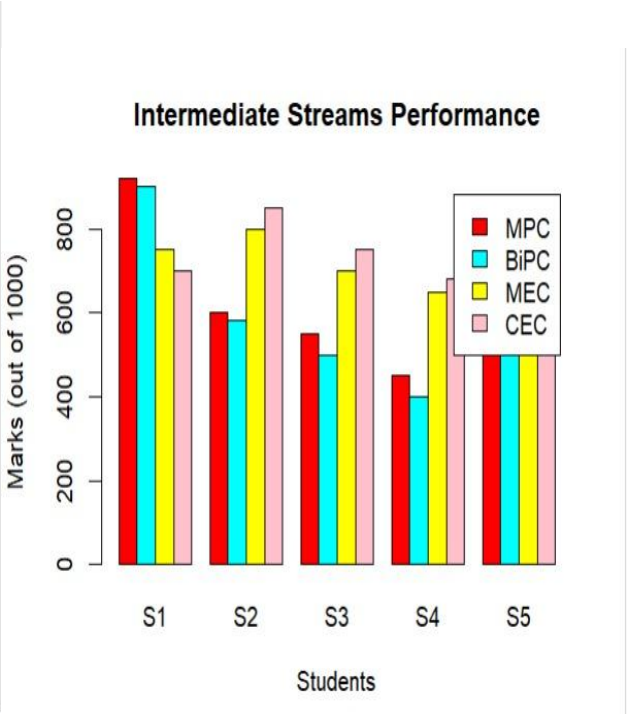
Students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BiPC
MEC
CEC
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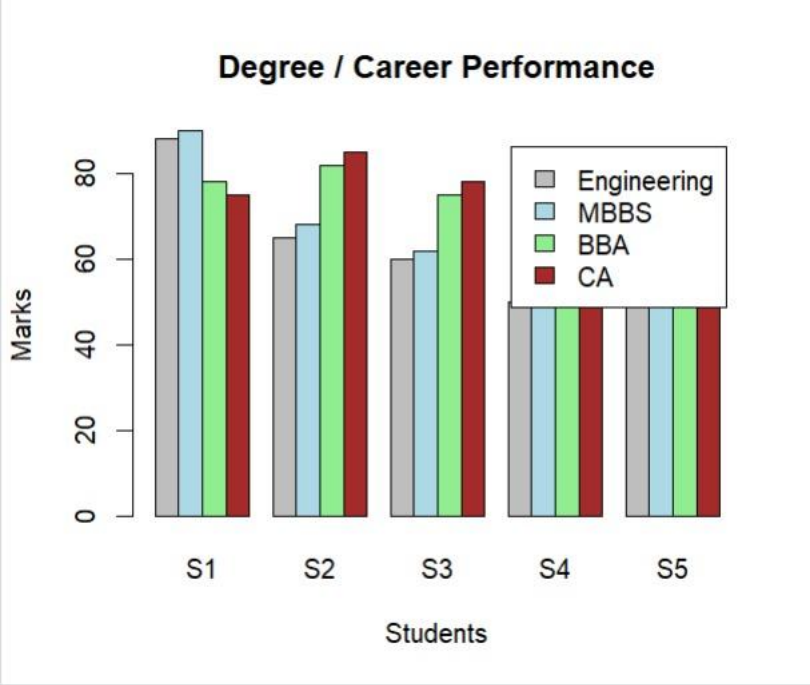
S2
S3
S4
S5
Students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40
60
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Engineering
MBBS
BBA
CA
S1
S2
S3
S4
S5
Students

```
Student: S1
10th Best Subject: Maths
Recommended Group: MPC

Student: S2
10th Best Subject: English
Recommended Group: Arts

Student: S3
10th Best Subject: Social
Recommended Group: MEC/CEC

Student: S4
10th Best Subject: Social
Recommended Group: MEC/CEC

Student: S5
10th Best Subject: Maths
Recommended Group: MPC
```

```
Student: S1
Best Stream: MPC
Recommended Degree: B.Tech

Student: S2
Best Stream: CEC
Recommended Degree: B.Com

Student: S3
Best Stream: CEC
Recommended Degree: B.Com

Student: S4
Best Stream: CEC
Recommended Degree: B.Com

Student: S5
Best Stream: MPC
Recommended Degree: B.Tech
```

```
Student: S1
Best Degree: MBBS
Marks: 90
Final Decision: Higher Studies (PG)

Student: S2
Best Degree: B.Com
Marks: 85
Final Decision: Higher Studies (PG)

Student: S3
Best Degree: B.Com
Marks: 78
Final Decision: Higher Studies (PG)

Student: S4
Best Degree: B.Com
Marks: 72
Final Decision: Job

Student: S5
Best Degree: MBBS
Marks: 86
Final Decision: Higher Studies (PG)
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